



STRATHMORE INSTITUTE OF MATHEMATICAL SCIENCES
MASTER OF SCIENCE IN DATA SCIENCE & ANALYTICS
ASSIGNMENT 1 -
DSA 8505: Bayesian Statistics

DATE: 27th Feb 2026

Instruction

- (a) Use python in this Assignment
- (b) Provide your commented scripts
- (c) Submit your answer through the Google Classroom by 23h59, 28th Mar 2025.

Objective:

This assignment aims to compare classical and Bayesian regression methods using real-world data. Students will explore the differences in parameter estimation, uncertainty quantification, and model performance.

Instructions:

1. **Dataset Selection:** Identify an appropriate dataset online (e.g., from Kaggle, UCI Machine Learning Repository, or government open data portals). The dataset must have multiple independent variables (predictors) and at least one continuous dependent variable.
2. **Data Preprocessing:** Load the dataset and perform exploratory data analysis (EDA). Handle missing values, outliers, and categorical variables appropriately. Split the data into training and testing sets. Classical Multiple
3. **Regression (Ordinary Least Squares - OLS):** Fit a multiple linear regression model using OLS. Check model assumptions (linearity, independence, homoscedasticity, normality). Report parameter estimates, R^2 , and residual analysis.
4. **Bayesian Multiple Regression:** Implement a Bayesian regression model using appropriate priors. Use probabilistic programming frameworks such as PyMC3, PyMC4, or Stan. Compare posterior distributions of parameters with classical estimates.

5. **Comparison and interpretation:** Compare the parameter estimates from OLS and Bayesian regression. Discuss differences in uncertainty quantification (credible vs. confidence intervals). Evaluate model performance using appropriate metrics (e.g., RMSE, MAE). Interpret findings in the context of real-world applications.

Submission Requirements:

- A Jupyter Notebook with well-commented code and results.
- A short report (2-3 pages) summarizing findings, interpretations, and comparisons.